# Stakeholder

**18+3 · seat empty**

N1 → N1.7 product path
Prefer nvidia/GR00T-N1.7-3B

Spine: Eagle-2 System 2 (~10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

H=16 · K=4 · N1 ≠ N1.7 · NOT a classical planner

NVIDIA · Fourier GR-1 · open humanoid VLA

## Club one-liner

**GR00T N1 = open dual-system humanoid VLA**

Eagle-2 System 2 (~10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

H=16 · K=4 · data pyramid · path to N1.7 ‖ $\pi_{0.5}$ on BEHAVIOR’26

**N1 ≠ N1.7 · NOT a classical planner**

All seats empty this session — still ship full five packs for club/GitHub

# Stakeholder value · open humanoid VLA path

### Product path

N1 paper → Isaac GR00T
→ Apache-2.0 N1.7 GA

### Challenge parity

BEHAVIOR’26 co-baseline
with $\pi_{0.5}$ — official

### Buy / build

Prefer finetune
nvidia/GR00T-N1.7-3B

### Data story

Pyramid + neural I2V
cuts teleop cost

### Bring-up

modality.json · emb tags
R1 Pro / NEW_EMBODIMENT

### Ops class

~16 GB infer / ~40 GB FT
(challenge freeze VL)

# Lineage · N1 → Isaac → N1.6 → N1.7 → BEHAVIOR’26

CLEAR: N1.7 GA = official co-baseline with $\pi_{0.5}$ · do NOT mix APIs/horizons

### N1
Mar 2025
Eagle-2 · 2.2B
H=16 · K=4

### Isaac
GR00T stack
LeRobot fmt
HF weights

### N1.6
BEHAVIOR1k
26.3% vs $\pi_{0.5}$
11.3% (README)

### N1.7
GA 3B
Cosmos-Reason2
H≤40 · Apache-2.0

### ’26
BEHAVIOR’26
∥ $\pi_{0.5}$
dual-track

### VERSION DISCIPLINE

Paper N1 ≠ N1.7 (backbone / horizon / package)
N1.6 BEHAVIOR1k numbers ≠ N1.7 forecasts

### π track (parallel)

$\pi_0$ → OpenPI → $\pi_{0.5}$ + RTC
BEHAVIOR’25 forks → ’26 co-baseline

Genealogy: SigLIP-2+SmolLM2 → Eagle-2 → N1 → N1.6 → N1.7 → BEHAVIOR’26 ∥ $\pi_{0.5}$

# VERSION DISCIPLINE · N1 ≠ N1.7

APIs / horizons / backbones differ — pin the checkpoint you serve

## Paper N1

GR00T-N1-2B · ~2.2B (1.34B VLM)

Eagle-2 · SigLIP-2 + SmolLM2

H=16 · K=4 · mid-layer 12

63.9 ms / chunk @ L40 bf16

Table 2/3 tattoo numbers live HERE

## N1.7 GA

nvidia/GR00T-N1.7-3B · ~3B

Cosmos-Reason2-2B (Qwen3-VL)

H ≤ 40 · new package / API

Apache-2.0 · BEHAVIOR’26 baseline

Do NOT quote N1 tables as N1.7

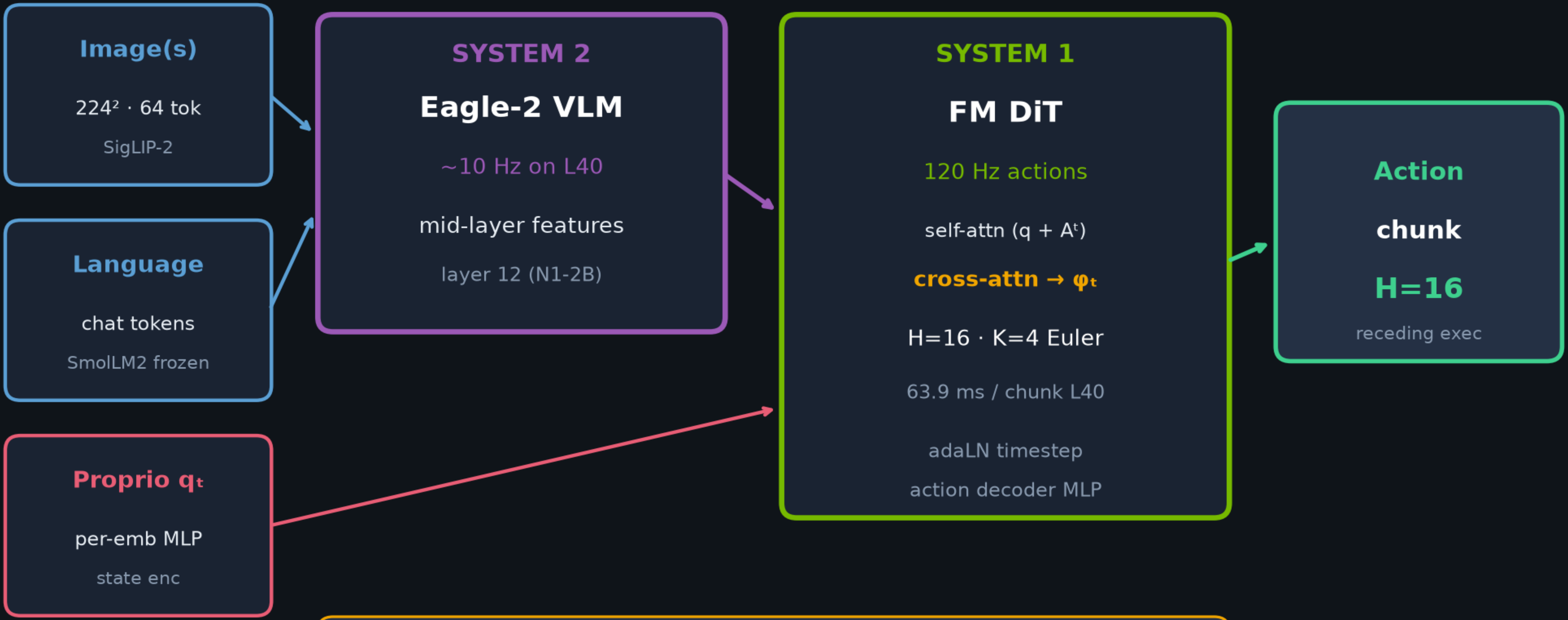
## Stakeholder / Visionary rule

Prefer finetune N1.7 for product/challenge · use N1-2B only for paper replication
Match served horizon to trained horizon · N1.6 BEHAVIOR1k ≠ N1.7 forecast

Also: N1 is NOT a classical planner — generative CFM chunk controller with dual-system packaging

# Dual-system VLA · Eagle-2 System 2 × FM DiT System 1

Jointly trained end-to-end · cross-attn bridge · NOT a classical planner




Cross-attn handoff = inspectable System-2 interface · foil to π fused MoE expert

# GR00T N1 vs $\pi_0$ / $\pi_{0.5}$ · same CFM family, different handoff

BEHAVIOR'26 ships BOTH as official baselines — dual-track is first-class

| AXIS | GR00T N1 | $\pi_0$ / $\pi_{0.5}$ |
|---|---|---|
| **Coupling** | Separate DiT · cross-attn | Action expert · shared self-attn |
| **VLM** | Eagle-2 (SmolLM2+SigLIP-2) | PaliGemma lineage |
| **Chunk / denoise** | H=16 · K=4 · 10/120 Hz | H=50 ($\pi_0$) · ~10 Euler · 20–50 Hz |
| **Embodiment I/O** | Per-emb MLPs · LAPA/IDM | Pad/mask shared dim |
| **Data bet** | Pyramid + neural I2V | OXE-scale + $\pi_{0.5}$ co-train |
| **Successor** | N1.7 ~3B · H≤40 · Apache-2.0 | $\pi_{0.5}$ + RTC + BEHAVIOR forks |

FLAG: modular dual-system / cross-attn DiT vs fused shared-attn MoE expert · same chunk+flow interface

# Data pyramid · quantity ↓ · embodiment-specificity ↑

~593M frames / ~8376 h · human + neural + DexMimicGen + real

**PEAK · Real robot**

GR-1 teleop ~88 h · OXE · AgiBot-Alpha · RH20T

**MIDDLE · Synthetic**

DexMimicGen ~780k traj · neural I2V 88h→827h · LAPA/IDM

**BASE · Web + human egocentric**

Ego4D · Ego-Exo4D · EPIC · HOI4D · Assembly-101 · LAPA latent actions

Visionary durable idea: open humanoid data pyramid survives backbone swaps (Eagle → Cosmos-Reason2 → ...)

# Tattoo numbers · remember these

Table 2 sim @100 demos · Table 3 real GR-1 · preserve DexMG mismatch

## SIM avg @100 (Table 2)

**N1-2B ~45%**

vs DP 33.4% · BC-T 26.4%

RoboCasa 32.1 · DexMG 66.5 · GR-1 50.0

App Table 4 DexMG@100 = 58.5 ≠ 66.5

## REAL GR-1 avg (Table 3)

**N1 full 76.8%**

vs DP full 46.4%

N1 @10% → 42.6% (≈ DP full −3.8 pp)

n≈10 trials · Pack Machinery n=5

### Systems

63.9 ms · H=16 · K=4 · L40

### Pretrain

~50k H100-h · BS 16384

### Neural

88h → 827h I2V · Defer factory

Empiricist: never quote Partial one-task smoke as Table 2 average

# Risks & costs · leadership must hear

### N1 ≠ N1.7 API

gr00t_n1d6 → n1d7 break
horizon / backbone change

### Short-horizon DNA

Paper = tabletop atomic
≠ house-scale BEHAVIOR

### Forgetting

Narrow FT erases
pretrained bimanual skills

### Isaac GPU needs

OmniGibson / no RT-core
sim stack is real cost

### Compute politics

50k H100-h pretrain
≠ student baseline

### Synthetic physics

Neural video MLLM-judge
≠ contact-rich correctness

# Success metrics · what “good” looks like

| | |
|---|---|
| **Paper suites** | Match/beat DP & BC-Transformer on RoboCasa / DexMG / GR-1 |
| **Challenge** | $\geq \pi_{0.5}$ starter on BEHAVIOR task progress / BDDL partial credit |
| **Serve** | Stable websocket + OmniGibson eval; horizon matched to train |
| **Data efficiency** | FT from ≤10% demos still competitive (paper real story) |
| **Version pin** | Documented ckpt ID · H · K · VL layer in every report |

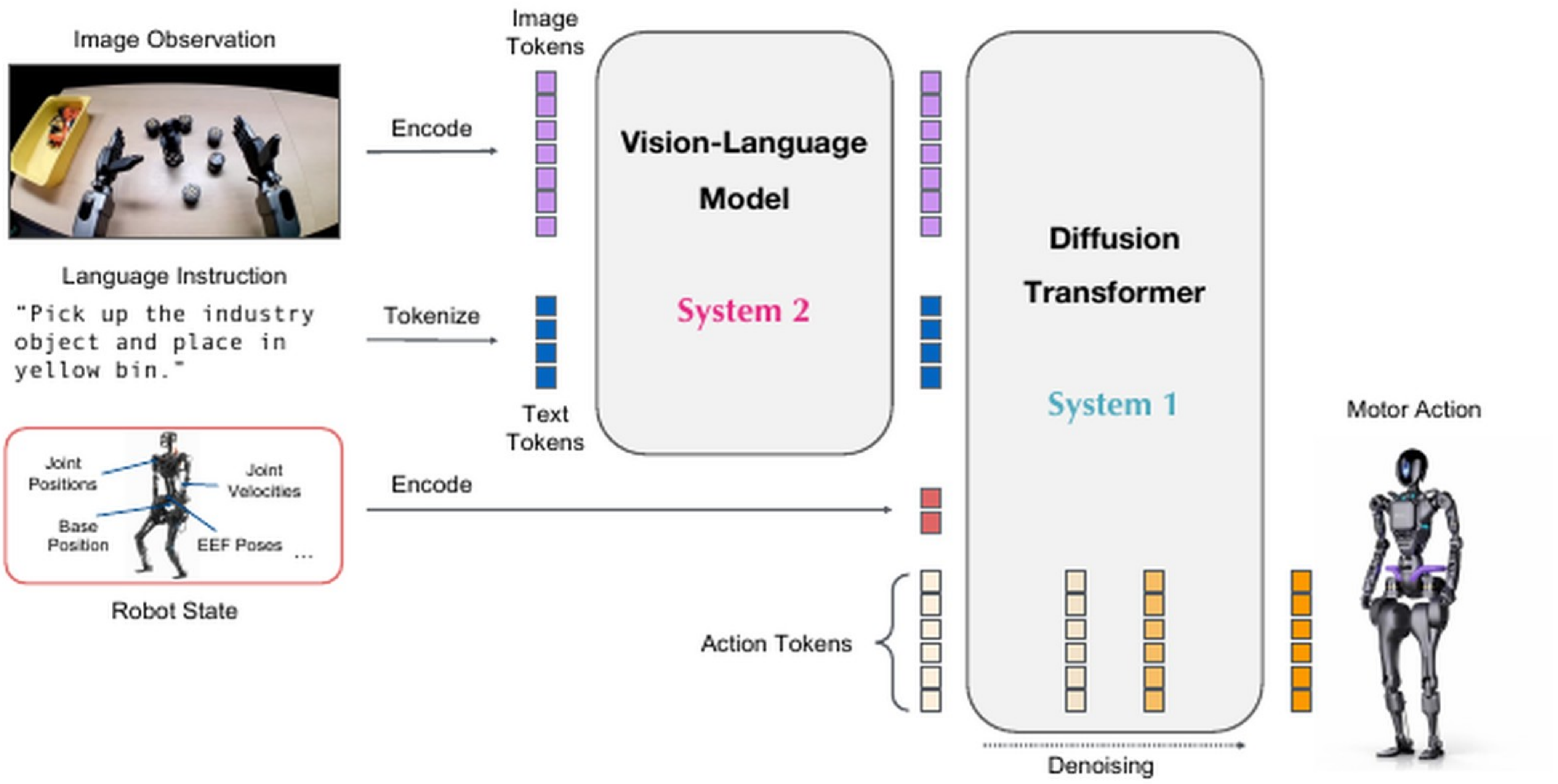



Figure 2: **GR00T N1 Model Overview.** Our model is a Vision Language Action (VLA) model that adopts a

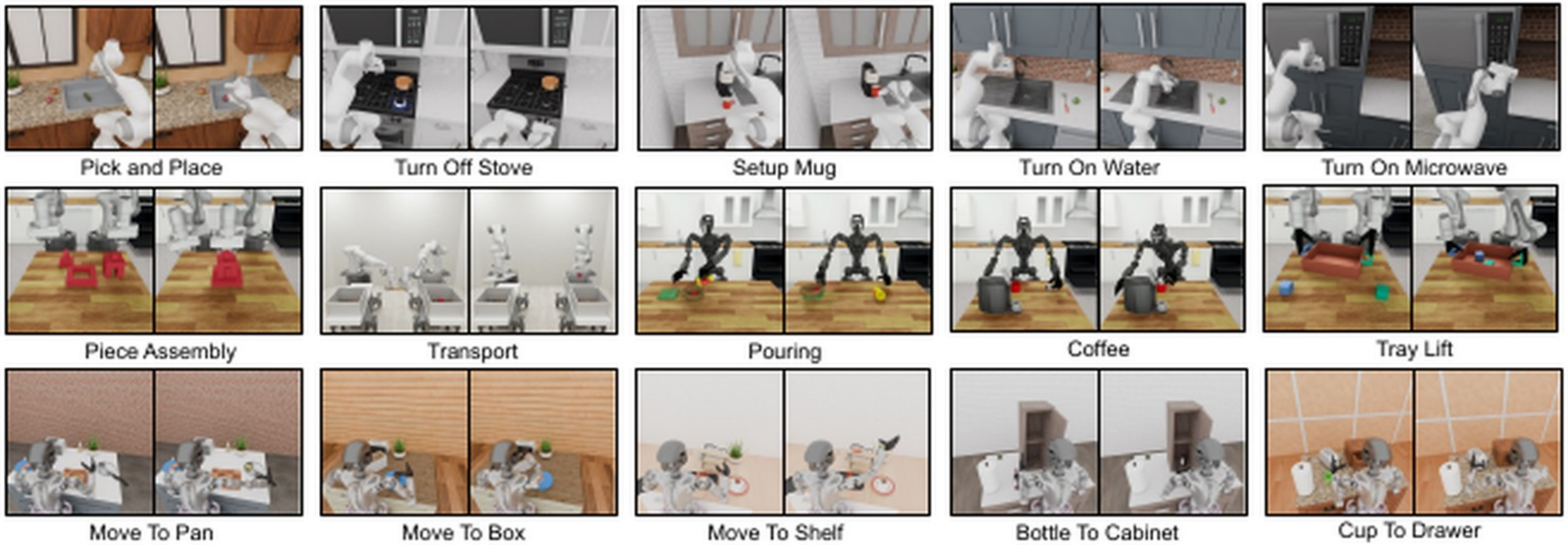



Figure 7: **Simulation Tasks.** Our simulation experiments use tasks from two open-source benchmarks (RoboCasa (Nasiriany et al., 2024) in the top row and DexMimicGen (Jiang et al., 2024) in the middle row) and a newly developed suite of tabletop manipulation tasks that closely resemble our real-world tasks (bottom row). We provide Omniverse renderings of the tasks above.

first-person perspectives of hand-object interactions (examples shown in Figure 14). Our video datasets include the following:

- **Ego4D** is a large-scale egocentric video dataset that includes diverse recordings of everyday activiillustrates some example tasks from these three benchmarks.

- **RoboCasa Kitchen (24 tasks, RoboCasa)**
  RoboCasa (Nasiriany et al., 2024) features a collection of tasks in simulated kitchen environments. We focus on 24 "atomic" tasks that involve foundational sensorimotor skills such as pick-and-place, door opening and closing, pressing buttons, turning faucets, and more. For each task, we use the publicly available dataset of 3000 demonstrations featuring the Franka Emika Panda arm, all generated with MimicGen (Mandlekar et al., 2023). The observation space includes three RGB images captured from cameras positioned on the left, right, and at the wrist. The state representation comprises the position and rotation of both the end-effector and the robot base, as well as the gripper's state. The action space is defined by the relative position and rotation of the end-effector along with the gripper state. We follow the same training and evaluation protocol outlined by Nasiriany et al. (2024).
- **DexMimicGen Cross-Embodiment Suite (9 tasks, DexMG)**
  DexMimicGen (Jiang et al., 2024) includes an array of nine bimanual dexterous manipulation tasks requiring precise two-arm coordination. Together, these tasks cover three bi-manual robot embodiments: (1) *Bimanual Panda Arms with Parallel-Jaw Grippers*: tasks include threading, piece assembly, and transport. The state/action space consists of the end-effector position and rotation of both arms, as well as the gripper states; (2) *Bimanual Panda Arms with Dexterous Hands*: tasks include box cleanup, drawer cleanup, and tray lifting. The state/action space consists of the end-effector position and rotation of both arms and hands; (3) *GR-1 Humanoid with Dexterous Hands*: tasks include pouring, coffee preparation, and can sorting. The state/action space consists of the joint position and rotation of both arms and hands, along with the waist and neck. We generate 1000 demonstrations for each task using the DexMimicGen data generation system and evaluate the model's ability to generalize to novel object configurations.

# Control spine · CFM cousin, dual-system packaging

NOT classical planner · generative continuous chunk controller

$$A_t^{\tau} = \tau A_t + (1-\tau)\varepsilon \quad \cdot \quad \mathcal{L}_{fm} = \mathbb{E}\|V_\theta - (\varepsilon - A_t)\|^2$$

**VL @ 10 Hz**

Eagle-2 mid features
cache across denoise

**Denoise K=4**

Euler on DiT
63.9 ms / H=16

**Commit + replan**

Don’t wait full H
open-loop; recede

Contrast: OpenPI action-expert FM · DP CNN/Transformer · π_RL PPO · Comet RFT

Same CFM math as $\pi_0$ — different reasoning↔motor handoff

# Stakeholder close

**Ship on N1.7 · pin versions · dual-track BEHAVIOR’26**

Prefer finetune nvidia/GR00T-N1.7-3B
Do not treat paper N1 APIs as N1.7

GR00T N1 · Oct 7 · all seats empty · still full five packs

# GR00T N1: An Open Foundation Model for Generalist Humanoid Robots

NVIDIA[1]


## Abstract

General-purpose robots need a versatile body and an intelligent mind. Recent advancements in humanoid robots have shown great promise as a hardware platform for building generalist autonomy in the human world. A robot foundation model, trained on massive and diverse data sources, is essential for enabling the robots to reason about novel situations, robustly handle real-world variability, and rapidly learn new tasks. To this end, we introduce GR00T N1, an open foundation model for humanoid robots. GR00T N1 is a Vision-Language-Action (VLA) model with a dual-system architecture. The vision-language module (System 2) interprets the environment through vision and language instructions. The subsequent diffusion transformer module (System 1) generates fluid motor actions in real time. Both modules are tightly coupled and jointly trained end-to-end. We train GR00T N1 with a heterogeneous mixture of real-robot trajectories, human videos, and synthetically generated datasets. We show that our generalist robot model GR00T N1 outperforms the state-of-the-art imitation learning baselines on standard simulation benchmarks across multiple robot embodiments. Furthermore, we deploy our model on the Fourier GR-1

**Prompt:** use the right hand to pick up cucumber to basket

**Prompt:** use the left hand to pick up spray bottle to basket

**Prompt:** use the left hand to pick up spray bottle to beige bowl

**Prompt:** pick up can from cutting board to pan

**Prompt:** pick up apple from cutting board to pan

**Prompt:** pick up tools from mesh cup to clear bin

**Prompt:** pick up the potato, place it into the microwave and close the microwave